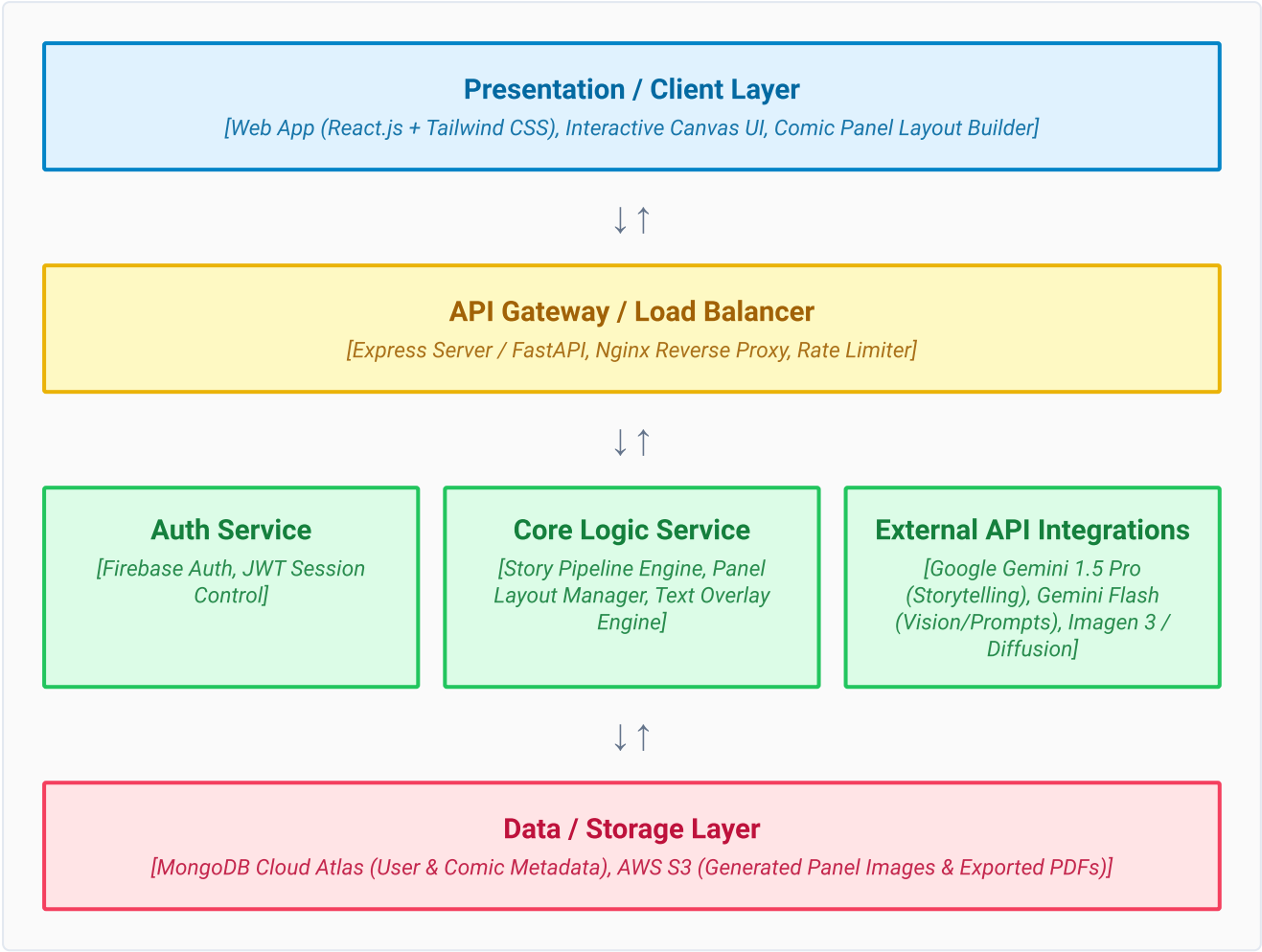


Solution Architecture

Date	15 March 2026
Team ID	SWTID-2026-9392
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the user interface, providing interactive story prompt input fields, comic panel customization tools, real-time preview, speech bubble overlay editing, and printable PDF export setup.	React.js Tailwind CSS HTML5 Canvas Lucide Icons
API Gateway	Handles incoming client HTTP requests, enforces API rate limits, validates user authorization tokens, and routes traffic efficiently to backend core logic and authentication endpoints.	Express.js Node.js Cors / Helmet Nginx
Authentication Service	Manages user registration, secure login sessions, OAuth2 integrations (Google Sign-In), and handles JWT token lifecycle for authorizing protected user comic projects.	Firebase Auth JSON Web Tokens OAuth 2.0
Core Logic Service	Executes comic generation workflows: breaks user narrative prompts into structured multi-panel scripts, formats prompts for Gemini AI, manages panel sequence consistency, and compiles generated artwork with dialogue.	Python / FastAPI LangChain Pillow (PIL) PDFKit
External API Integrations	Interactions with AI foundation models. Uses Google Gemini 1.5 Pro to script comic scenes, dialogues, and visual prompts, and Imagen 3 / Diffusion models for high-quality panel image generation.	Google Gemini 1.5 Pro Gemini Flash Imagen 3 API
Database & Object Storage	Persists user profiles, comic metadata, panel structure templates, and dialogue scripts in Document DB. Stores high-resolution panel images and final exported PDF files in cloud object storage.	MongoDB Atlas AWS S3 Mongoose ORM

